Backtest statistics: calculation guide

Scope

2Y futures backtests, 4 January 2016 through 31 December 2025.
Statistics are calculated separately for each strategy and reporting period.
Daily strategy returns reflect the position and futures return in the saved daily CSV.

Win ratio

Win ratio (%) = 100 x positive-return days / nonzero-return days.
Zero-return days, including inactive days, are excluded from the denominator.
If all days have zero returns, the win ratio is undefined (blank).

Risk-reward ratio

Risk-reward = average positive daily return / absolute average negative daily return.
Zero-return days are excluded from both averages. The ratio is not based on trades.
If there are no positive days or no negative days, the ratio is undefined (blank).

Example

Six daily returns: +0.2%, +0.1%, +0.3%, -0.1%, -0.3%, 0%.
Win ratio = 3 / 5 = 60%. Average positive return = 0.2%.
Average loss magnitude = 0.2%; risk-reward ratio = 0.2 / 0.2 = 1.0.

Calmar ratio

Calmar = arithmetic annualized return / absolute maximum drawdown.
Annualized return is the mean daily return multiplied by 252, matching the existing statistics. If maximum drawdown is zero, Calmar is undefined (blank).

Maximum drawdown duration

Maximum duration is the longest time in calendar days from a running wealth peak to the first observation at or above that peak. If unrecovered, it ends at the final observation. Each annual period starts a new wealth and drawdown series.
The longest episode need not contain the deepest drawdown.

Interpretation

These are daily-return statistics, not trade win rates or per-trade reward/risk.
They use the recorded backtest returns, without transaction costs.